

Develop a fully advanced and interactive **Smart Hospital Management System** using **React only**. The application should simulate a real-world healthcare management platform with modern UI/UX, dynamic dashboards, patient management, doctor scheduling, analytics, and advanced frontend architecture.

Project Requirements

1. Authentication & User Roles

- Login & Signup Pages
- Role-Based Access:
 - Admin
 - Doctor
 - Receptionist
 - Patient
- Protected Routes
- Form Validation
- Session Management UI

2. Smart Admin Dashboard

Create an enterprise-level dashboard displaying:

- Total Patients
- Available Doctors
- Appointments Overview
- Emergency Cases
- Revenue Analytics
- Bed Availability
- Hospital Activity Feed
- Daily Reports
- Interactive Charts & Statistics

3. Patient Management Module

- Add/Edit/Delete Patients
- Patient Medical History
- Prescription Records

- Patient Profile Management
- Search & Filter Patients
- Emergency Contact Information
- Admission & Discharge Tracking

4. Doctor Management Module

- Doctor Profiles
- Department Allocation
- Doctor Availability Status
- Appointment Scheduling
- Doctor Performance Dashboard
- Consultation History

5. Appointment Booking System

- Book Appointments
- Appointment Approval Workflow
- Calendar Scheduling Interface
- Time Slot Management
- Appointment Notifications
- Upcoming Appointments Dashboard

6. Pharmacy & Billing Module

- Medicine Inventory UI
- Billing Dashboard
- Payment Status Tracking
- Invoice Generation UI
- Revenue Reports
- Expense Analytics

7. AI Health Analytics Dashboard

- Patient Health Monitoring UI
- Disease Statistics Visualization
- Health Risk Indicators

- Emergency Alerts Dashboard
- Recovery Progress Tracking
- Smart Recommendation Panel

UI/UX Requirements

- Modern Healthcare Dashboard Design
- Fully Responsive Layout
- Collapsible Sidebar Navigation
- Dark/Light Mode
- Animated Dashboard Components
- Responsive Data Tables
- Interactive Cards and Charts
- Smooth Page Transitions
- Professional Form Designs
- Popup Modals & Notifications

Technical Requirements

- Use React Functional Components
- Use React Hooks:
 - useState
 - useEffect
 - useContext
 - useReducer
 - useMemo
- React Router for Navigation
- Context API for Global State Management
- Reusable Component Architecture
- Dynamic Rendering
- Lazy Loading & Code Splitting
- Custom Hooks
- Local Storage Integration

- Mock API Handling
- Error Boundaries
- Responsive Grid Layouts

Additional Advanced Features

- Real-Time Notification Center
- Calendar & Scheduler Widgets
- Advanced Search & Filters
- Multi-Step Registration Forms
- Dashboard Widget Customization
- Loading Skeleton Screens
- Toast Notifications
- Infinite Scroll Patient Records
- Activity Logs & Audit Tracking

Expected Outcome

The final application should function like a real-world enterprise hospital management platform with advanced React architecture, professional dashboards, dynamic frontend functionality, reusable components, responsive layouts, and scalable user interface design suitable for production-level healthcare systems.